Windows Server Fundamentals
&
Computer Hardware

Windows Server Fundamentals

- Windows Server Fundamentals
  - Network Basics
  - Defining Server roles
  - Hardware
  - Virtual Server
- Servers Information: Definition and Purpose
  - Active Directory
  - DNS Server
  - DHCP Server
  - Web Server
  - FTP Server
- Computer Hardware
  - What is computer memory?
  - Cache memory
  - Common memory errors

Labs

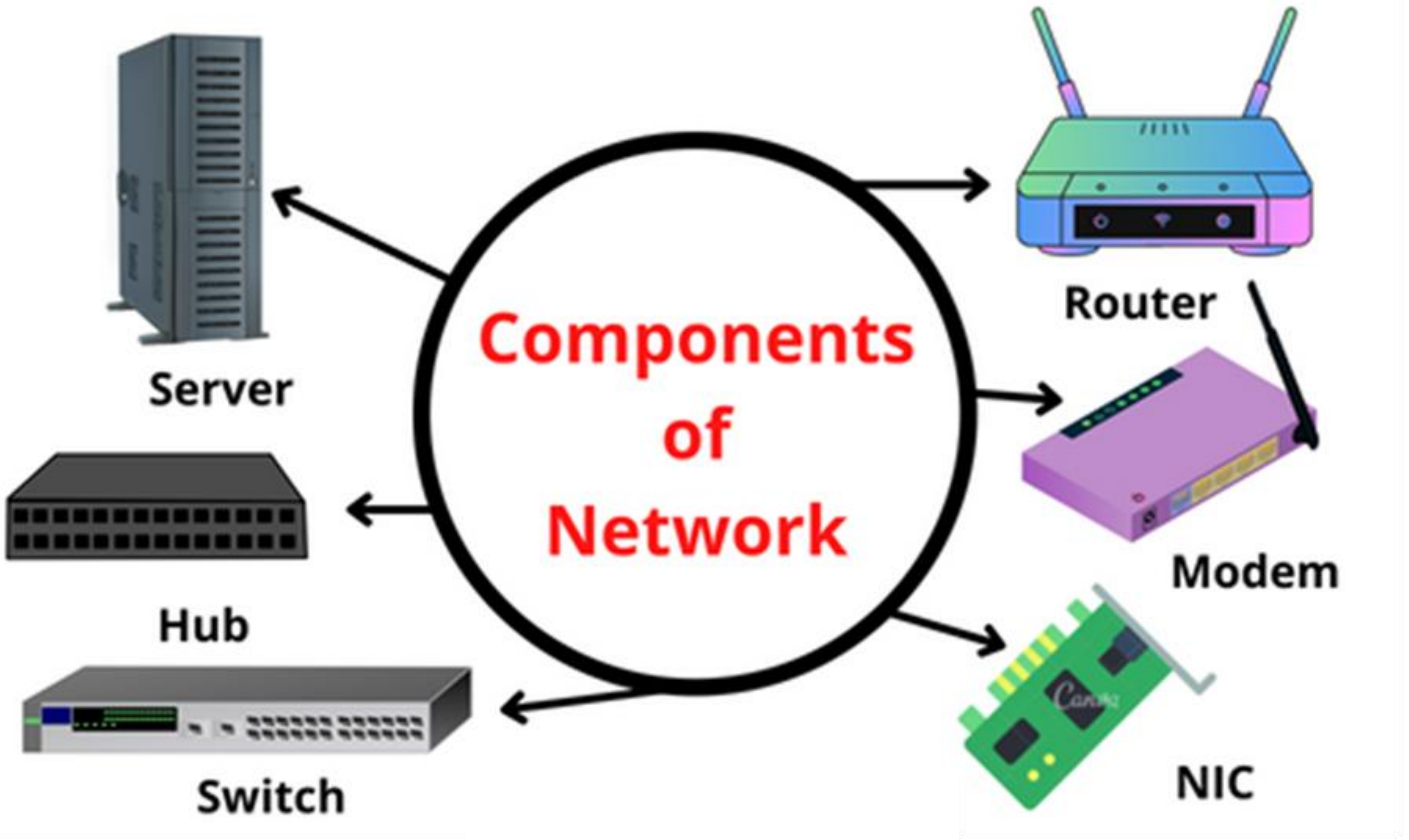
- Students will **create hard disk partitions**, setup services in Windows Server 2022.
- Students will use **msinfo32** utility and check the results from their systems
- Students will analyze the given **msinfo32** output and answer the questions based on the same

Network Basics

- A computer network is a system that connects many independent computers to **share** information (data) and resources.
- A computer network is made up of two main parts:
  - **Devices** (called nodes) &
  - **Connections** (called links)
- The **links** connect the devices to each other.
- The rules for how these connections send information are called communication **protocols**.
- The starting and ending points of these communications are often called **ports**.

Network devices

- ✓Network Interface Card (NIC)
- ✓Repeater
- ✓Hub
- ✓Bridges
- ✓Switches
- ✓Routers
- ✓Gateways



IP Address

- An IP address (Internet Protocol address) is a unique numerical identifier assigned to every device connected to a computer network.
- It is used to identify and locate devices and to route data over the network.
- Main Functions of an IP Address
  - *Identification* – Identifies a particular device
  - *Location addressing* – Shows the device's network
  - *Communication* – Helps send and receive data
- Types of IP Addresses:
  - *IPv4* – 32-bit address
  - *IPv6* – 128-bit address

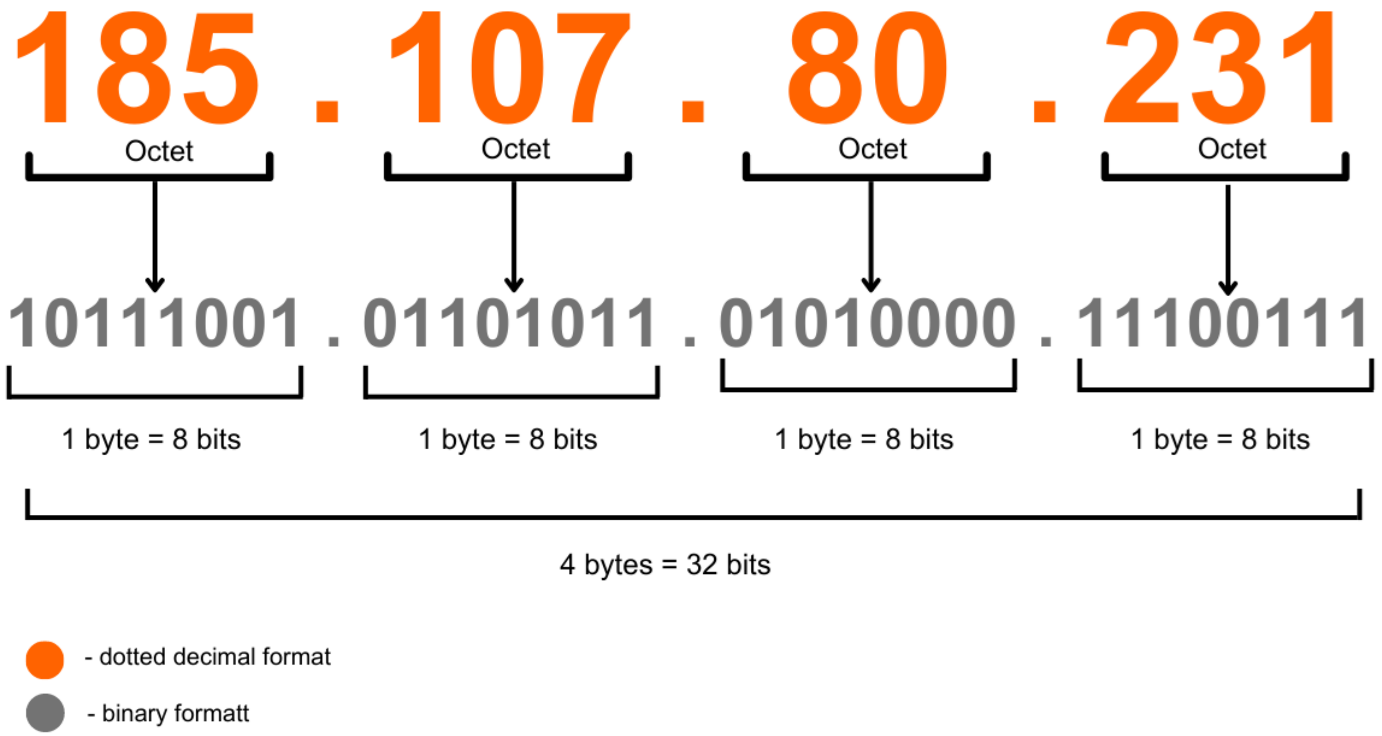
IPv4 Addresses

- IPv4 is the older and more widely used IP addressing format.
- An IPv4 address is a 32-bit number divided into four 8-bit sections (octets), each separated by a dot.
- Each octet can hold a value between 0 and 255, resulting in an address structure like 192.168.1.1.

```
IPv4 Address: 192.168.0.1
Binary Format: 11000000.10101000.00000000.00000001
```

IPv4 Addresses

- Internet Protocol Address.
- It is 32 Bits address.
- It’s a logical address.
- IP must be unique & universal.
- 2 ways to display an IP address:
  - ✓ Binary Notation
  - ✓ Dotted-Decimal Notation



IP ADDRESS Classes

- Class A (**1.0.0.0 to 126.0.0.0**): Supports large networks with millions of hosts.
- Class B (**128.0.0.0 to 191.255.0.0**): Supports medium-sized networks.
- Class C (**192.0.0.0 to 223.255.255.0**): Ideal for small networks.
- Class D (**224.0.0.0 to 239.255.255.255**): Reserved for multicast.
- Class E (**240.0.0.0 to 255.255.255.255**): Reserved for experimental or research purposes.

Private IP Address Ranges

- The private address ranges set by the Internet Engineering Task Force (IETF) for IPv4 are:

Class	Starting Address	Ending Address	Default Subnet Mask	Usage	Special Notes
A	0.0.0.0	127.255.255.255	255.0.0.0	Large networks	Includes `127.0.0.0/8` (reserved for loopback)
B	128.0.0.0	191.255.255.255	255.255.0.0	Medium-sized networks	N/A
C	192.0.0.0	223.255.255.255	255.255.255.0	Small networks	Commonly used for local area networks (LANs)
D	224.0.0.0	239.255.255.255	N/A	Multicasting	Used for sending data to multiple hosts simultaneously
E	240.0.0.0	255.255.255.255	N/A	Reserved for experimental purposes	Not used for general communication

Public and Private IP Addresses

Public IP Addresses

- are globally unique and accessible over the internet.
- They are assigned by the Internet Assigned Numbers Authority (IANA) and are used to identify devices on a wide-scale network, such as the internet.

Private IP Addresses

- are used within private networks (e.g., home or office networks).
- These addresses are not unique globally and are meant for devices within a single network to communicate without directly connecting to the internet.

IPv4 Addresses

4,294,967,296

IPv6 Addresses

340,282,366,920,938,463,463,374,607,431,768,211,456

Server VS Client

- **Client Operating System**: A Client Operating System is designed for end users to perform everyday tasks on personal devices.
  - Examples:
    - Windows 10 / Windows 11
    - MacOS, Ubuntu Desktop
    - Android / iOS
  - Used for:
    - Web browsing
    - Office work
    - Media, Gaming
    - Programming (personal use)

Server VS Client

- **Server Operating System**: A Server Operating System is designed to manage, serve, and control network resources for multiple users or other computers (clients).
- Examples:
  - Windows Server 2019 / 2022
  - Red Hat Enterprise Linux (RHEL)
  - Ubuntu Server
  - CentOS
- Used for:
  - Hosting websites, File sharing
  - Database services, Email services
  - User authentication (Active Directory)
  - Virtualization, Application hosting

Server Role

- Server roles are the primary functions or services that a server is configured to perform in a network.
- List of Windows Server Roles:
  - AD DS, AD CS, AD FS,
  - DHCP, DNS
  - File Services, Print Services
  - IIS, Hyper-V
  - RDS, WDS, WSUS

Active Directory (AD)

- Active Directory is used to centrally manage users, computers, and security in a Windows network environment.
- Active Directory helps an organization to:
  - Create and manage user accounts
  - Manage computer accounts
  - Apply security policies (Group Policies)
  - Control login access across the network
  - Manage printers, shared folders, and applications
  - Provide authentication and authorization

Components of AD

- Active Directory
  - Physical Components
    - RODC
    - GC
    - DC
    - Sites
  - Logical Components
    - Domain
    - Organizational Unit (OU)
    - Tree / Forest
    - Schema

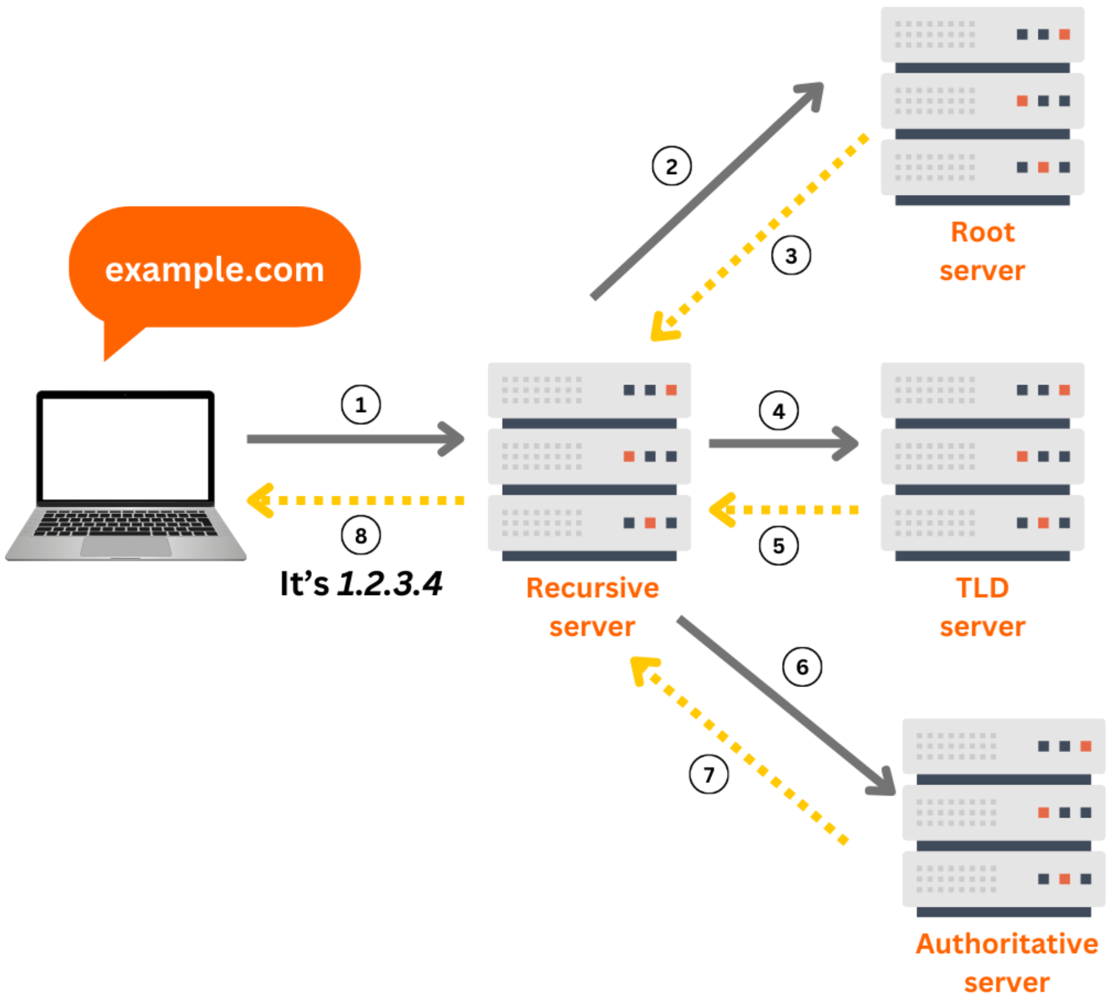
DNS Server

- DNS (Domain Name System) is like the phonebook of the Internet.
- It converts human-readable domain names (like www.google.com) into machine-readable IP addresses (like 1.1.1.1) so computers can find each other on a network.
- DNS allows:
  - Easy website access by name
  - Email routing
  - Load balancing
  - Security checks (filtering, blocking)
  - Directory services like Active Directory to function

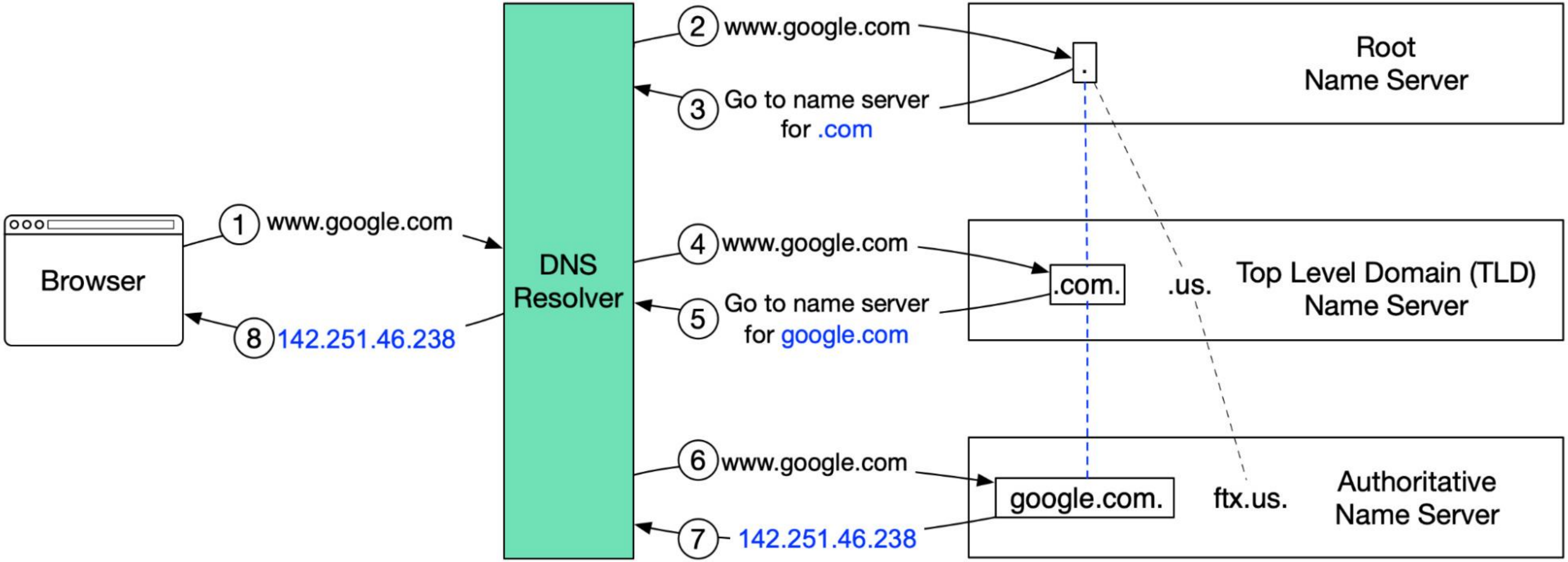
Types of DNS Servers

DNS Type	Function
Recursive Resolver	Receives request from client and finds the answer
Root Name Server	Directs the resolver to TLD servers
TLD Server	Manages .com, .org, .net, etc.
Authoritative Server	Stores actual IP of the domain

Working of DNS Server



Working of DNS Server



DHCP Server

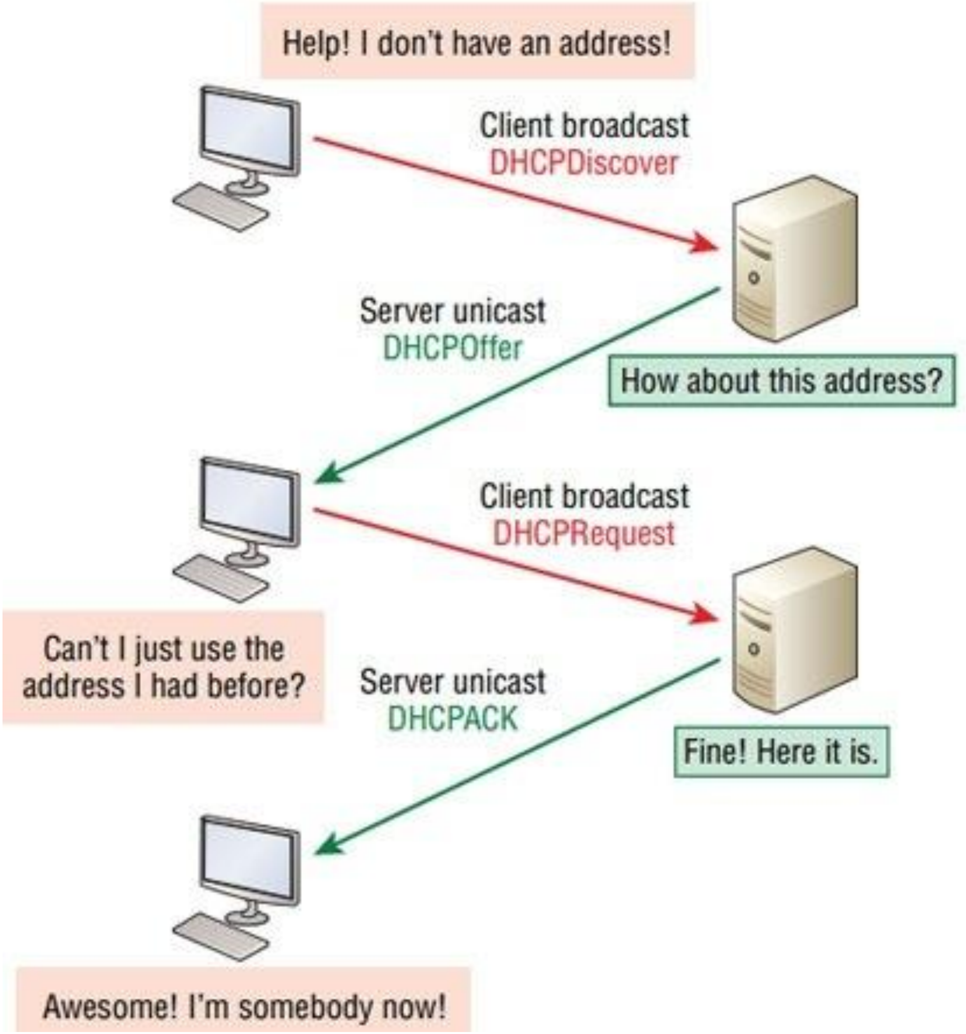
- A DHCP server (Dynamic Host Configuration Protocol) is a network service that automatically assigns IP addresses and other network configuration details to devices (clients) on a network.
- It saves time, reduces errors, and ensures smooth network communication.
- Types of DHCP Implementation
  - *Windows DHCP Server* (Windows Server role)
  - *Linux DHCP server* (ISC-DHCP, Dnsmasq)
  - *Router DHCP* (Home/Office routers)
  - *Cloud DHCP* (Azure/AWS VPC)

What Does a DHCP Server Provide?

- A DHCP server automatically provides:
  - IP address
  - Subnet mask
  - Default gateway
  - DNS server address
  - Lease time (how long the IP is valid)
- These details allow a device to connect and communicate on the network and the internet.

Working of DORA process

- A new device connects to the network
- It sends **DHCP Discover** message
- DHCP server responds with **DHCP Offer**
- The device sends **DHCP Request**
- DHCP server sends **DHCP Acknowledge**
- Device receives IP & network settings



Important DHCP Concepts

Term	Meaning
Scope	Range of IPs that can be assigned
Lease time	Duration of IP given
Reservation	Same IP to same device (MAC-based)
Exclusion range	IPs not given by DHCP
DHCP Relay Agent	For different subnets/VLANs
APIPA	169.254.x.x if DHCP fails

Web Server

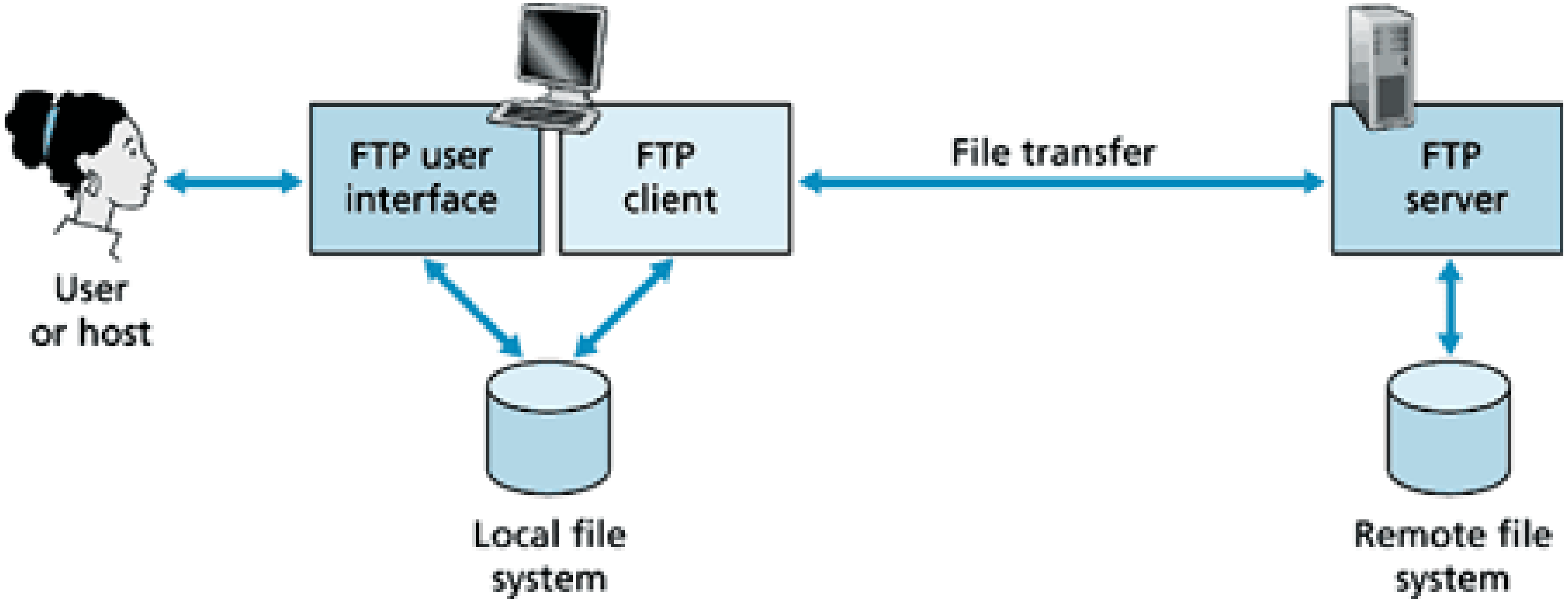
- A web server is a computer system or software that stores, processes, and delivers web pages to users over the Internet or an internal network using HTTP/HTTPS protocol.
- Types of Web Servers (Examples)
  - **Apache** → Linux / Windows
  - **NGINX** → High-performance
  - **Microsoft IIS** → Windows Server
  - **LiteSpeed** → Hosting
  - **Tomcat** → Java apps

FTP Server

- An FTP (File Transfer Protocol) Server is a system or software used to store and transfer files between computers over a network or the Internet.
- It allows users to:
  - Upload files to a server
  - Download files from a server
  - Manage files remotely (create, delete, rename)
- Types of FTP Modes
  - Active Mode
  - Passive Mode (most common)

Common FTP Server Software

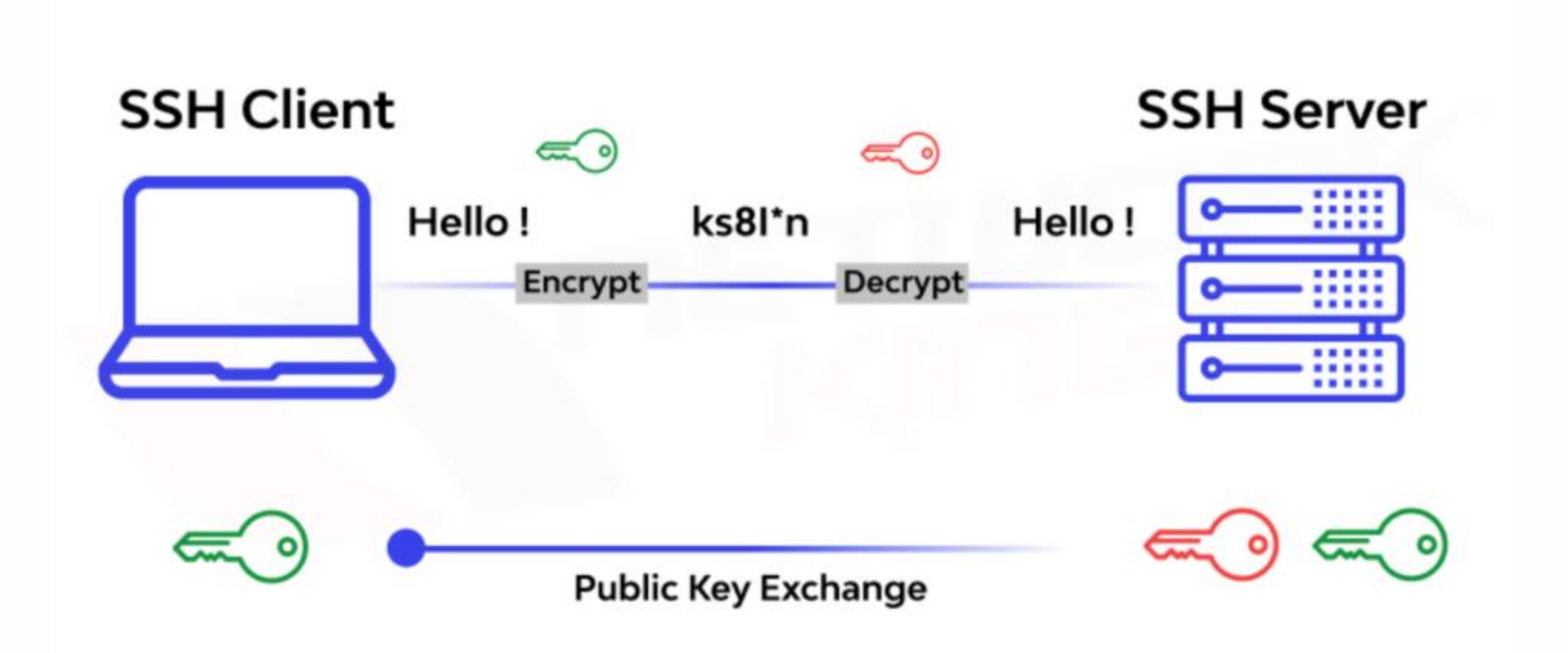
- FileZilla Server
- vsftpd (Linux)
- ProFTPD (Linux)
- IIS FTP (Windows Server)



SSH Server

- An SSH (Secure Shell) Server is a service that allows users to securely connect and control a remote computer over a network using encrypted communication.
- SSH is used to:
  - Securely access remote servers
  - Run commands on a remote system
  - Transfer files securely (SCP / SFTP)
  - Perform server administration
  - Create encrypted tunnels
- Unlike Telnet, SSH encrypts all data to prevent hacking and spying.

SSH Architecture



Computer Hardware

Computer Hardware

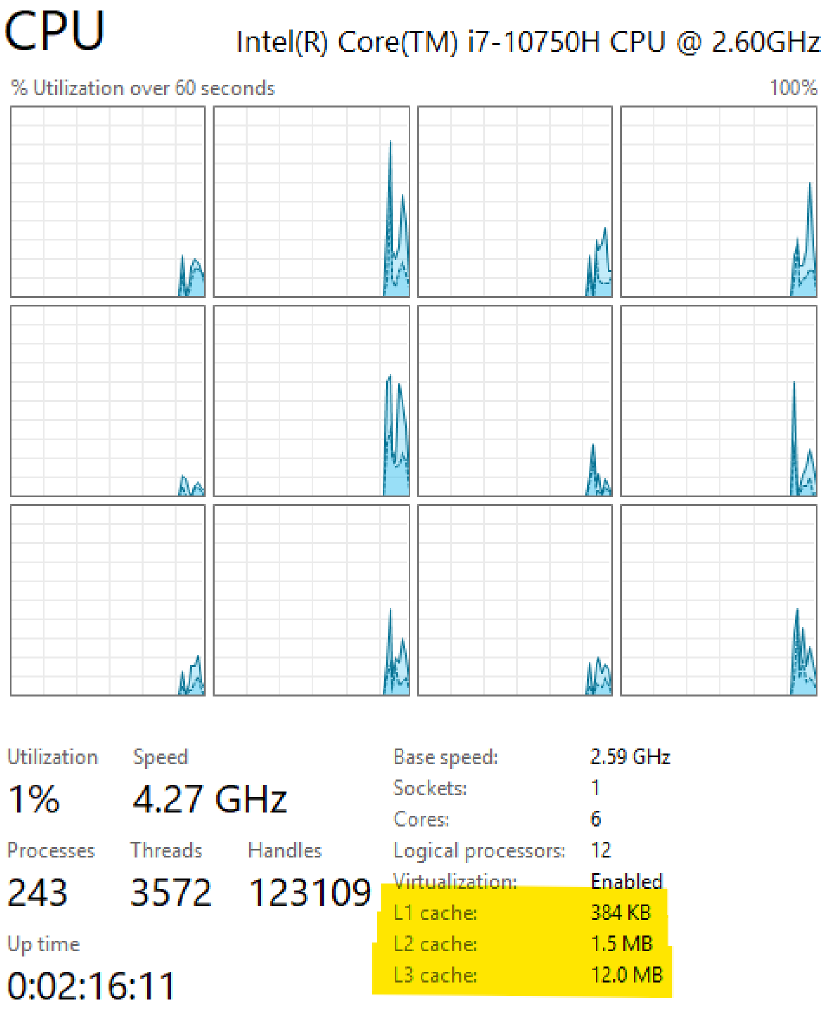
- What is computer memory?
- Cache memory
- Common memory errors

Computer Memory

- Computer memory is the part of a computer system that is used to store data, instructions, and programs so the CPU can access and use them to perform tasks.
- Types of Computer Memory:
  - **Primary Memory** (Main Memory)
    - Directly accessed by the CPU
    - Very fast
    - Used to store data currently in use
    - Types:
      - RAM (Random Access Memory) – Temporary, lost when power is off
      - ROM (Read Only Memory) – Permanent, stores startup instructions
  - **Secondary Memory** (Storage)
    - Used for long-term storage
    - Slower than primary memory
    - Examples:
      - Hard Disk (HDD)
      - Solid State Drive (SSD)
      - USB drives, Memory cards

Cache Memory

- Cache memory is a small, extremely fast type of memory located very close to the CPU (often inside the processor chip).
- Its main purpose is to store frequently used data and instructions so the processor can access them much faster than from RAM.
- This helps increase the overall speed and performance of a computer.
- Cache Memory reduces:
    - ✓ CPU wait time
    - ✓ Execution time of programs
    - ✓ System lag



Levels of Cache Memory

Level	Full Form	Location	Speed	Size
L1 Cache	Level 1	Inside CPU Core	Fastest	Smallest
L2 Cache	Level 2	Near CPU / On chip	Very fast	Medium
L3 Cache	Level 3	Shared among cores	Fast	Largest (of cache)

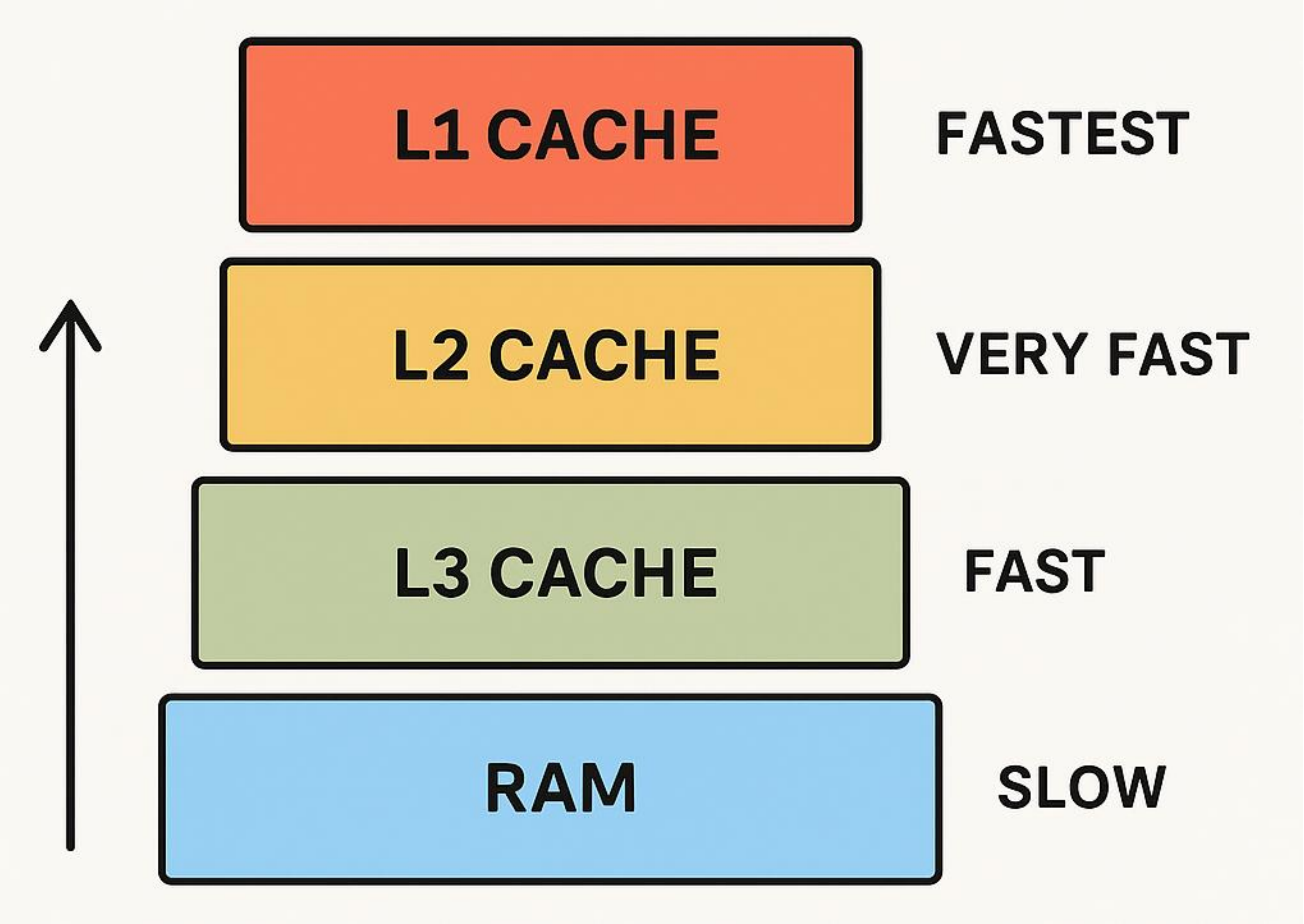
Advantages & Disadvantages

Advantages

- ✅ Faster system performance
- ✅ Reduces latency
- ✅ Efficient CPU utilization

Disadvantages

- ❌ More expensive
- ❌ Limited capacity



That's all Folks!